Cardiac Organoid Platform with Micropillar-Assisted Calcium and Optical Signal Analysis

Young Hyun Kim[1], Keel Yong Lee[1,2]*

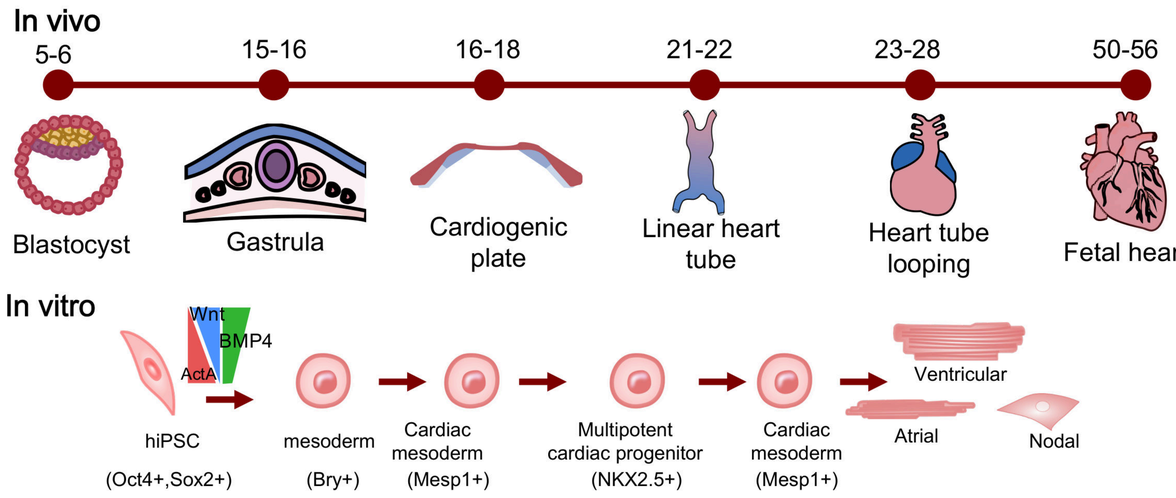
[1] Department of Integrative Bioscience and Biotechnology, Sejong University, Seoul, Korea

[2]Institute of Bioscience and Biotechnology, Sejong University, Seoul, Republic of Korea

ABSTRACT

The cardiac organoid platform integrates micropillar-assisted culture to achieve consistent organoid placement, improved optical access, and minimized motion artifacts during imaging. Combining calcium imaging and intrinsic optical signal analysis, the system captures lineage-dependent differences in calcium handling and electrophysiological organization. Importantly, the micropillar-assisted configuration enhances experimental reproducibility and **enables high-throughput functional screening** by allowing multiple organoids to be analyzed in parallel with standardized geometry and imaging conditions.

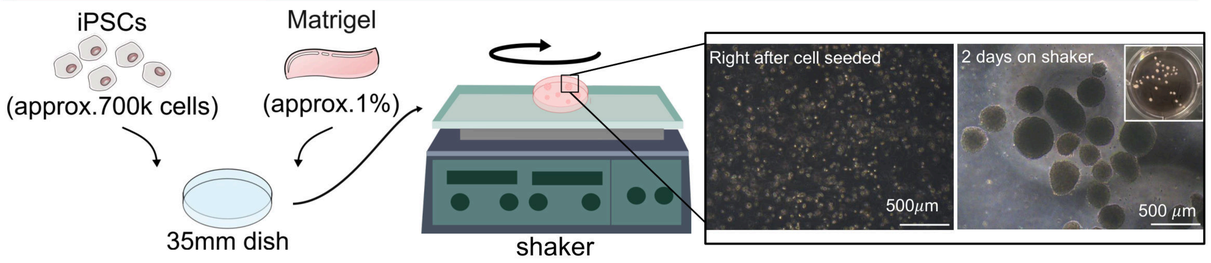
BACKGROUND



Cardiac organoids are widely used to study human cardiogenesis, making reliable structural and functional evaluation essential despite the challenges posed by their heterogeneous morphology and spontaneous motion

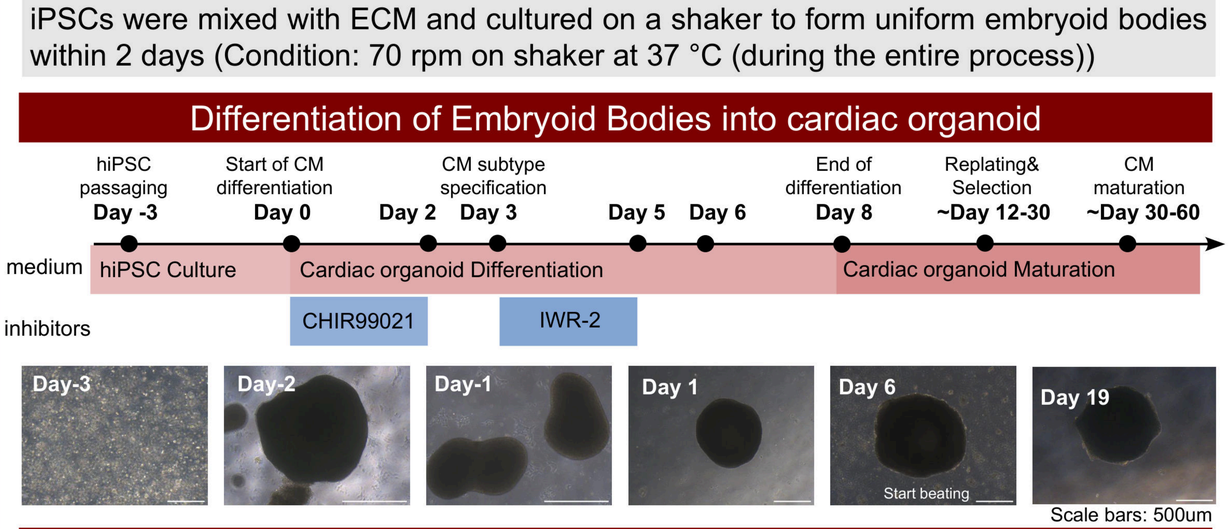
METHOD

Generation of cardiac organoid with shaker

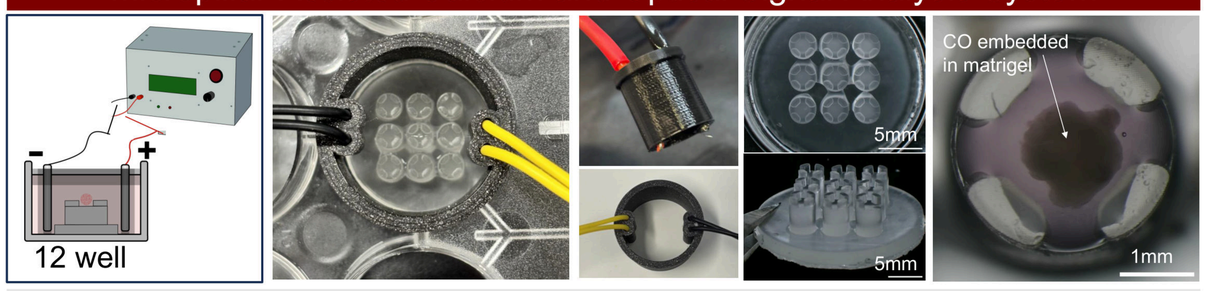


iPSCs were mixed with ECM and cultured on a shaker to form uniform embryoid bodies within 2 days (Condition: 70 rpm on shaker at 37 °C (during the entire process))

Differentiation of Embryoid Bodies into cardiac organoid



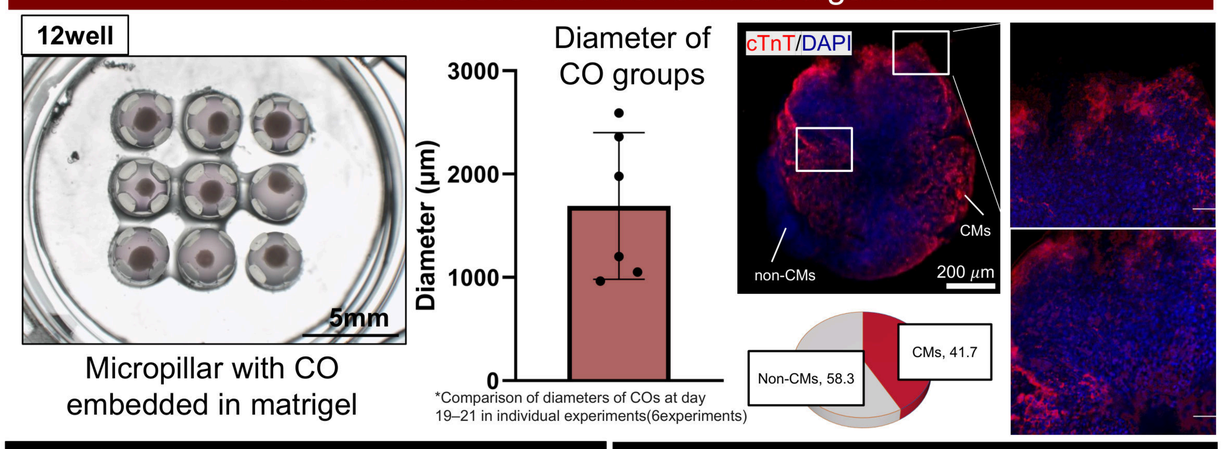
Micropillar-Assisted Calcium and Optical Signal Analysis System

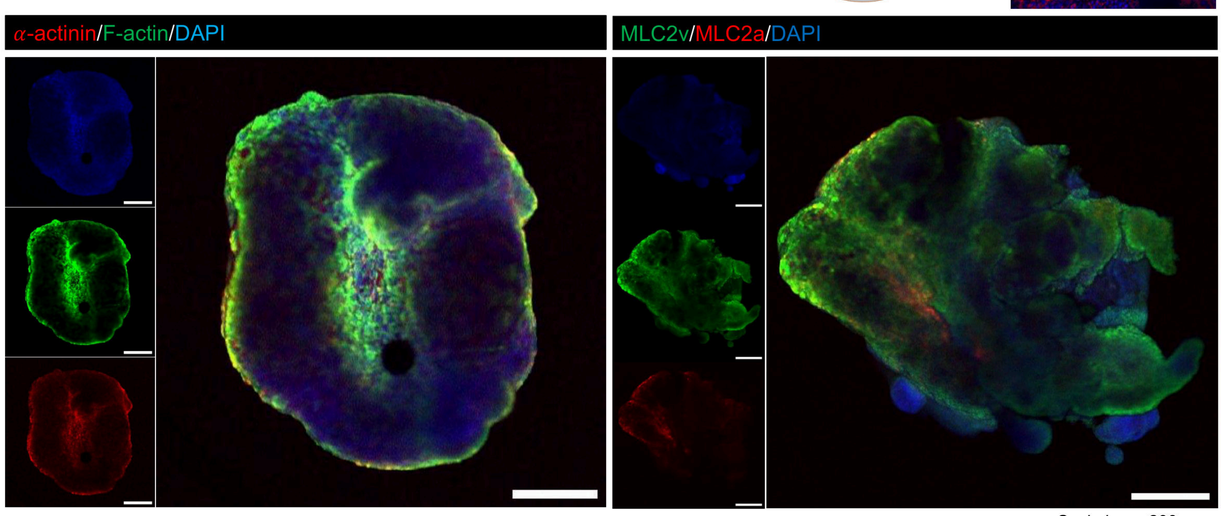


SLA-printed micropillar array designed to fit a 12-well plate and a PLA-based electrode setup used for calcium and intrinsic optical signal measurements

RESULTS

Structural Assessment of Cardiac organoids

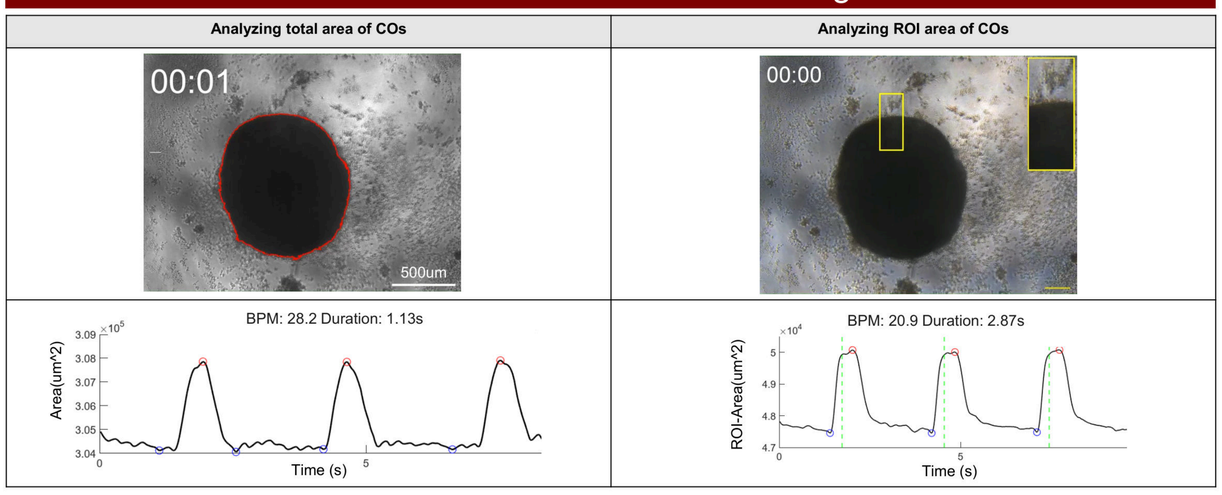




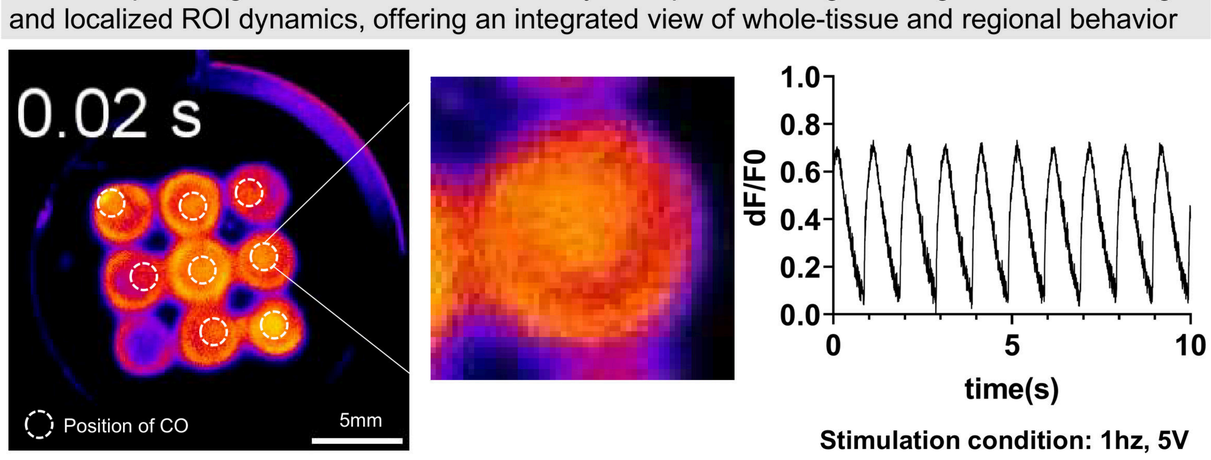
Scale bars: 200 μm

Identification of CMs and atrium/ventricle-like areas in CF-hHOs

Functional Assessment of Cardiac organoids



Intrinsic optical signal-based mechanical analysis captures both global organoid area changes and localized ROI dynamics, offering an integrated view of whole-tissue and regional behavior



Ca $^{2+}$ signaling of Cardiac organoids. By measuring the distance and time between peaks, conduction velocity can be determined)

Further Study

Chamber-specific cardiac organoids will be generated to establish disease-specific in vitro models that capture atrial- and ventricular-dependent pathophysiology. These models will be evaluated using both electrophysiological and molecular analyses to define chamber-specific disease signatures more precisely.

ACKNOWLEDGMENTS

This research was supported by a grant of the Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health & Welfare, Republic of Korea (No. RS-2025-24536077), and by a grant from the National Research Foundation of Korea (NRF) funded by the Korea government (MSIT) (No. RS-2025-23323900)

REFERENCE

[1] Lee, Seul-Gi, et al. "Generation of human iPSCs derived heart organoids structurally and functionally similar to heart." *Biomaterials* 290 (2022): 121860.

[2] Kleinsorge, Mandy, and Lukas Cyganek. "Subtype-directed differentiation of human iPSCs into atrial and ventricular cardiomyocytes." *STAR protocols* 1.1 (2020): 100026.